LOTQ-0041 / DATA CLEANING / INTERMEDIATE / FIELD GUIDE

Missing Data, Outliers, and Bad Ticks

Separate valid market tails from defects and repair anomalies without inventing history.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0041
CATEGORY	Data Cleaning
DIFFICULTY	Intermediate
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

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How to use this manual

INTERMEDIATE

1. DEFINE EXPECTED EVIDENCE

Start with instrument, field, grain, calendar, source, and decision cutoff. An empty cell becomes meaningful only after the system proves that an observation was expected.

2. SEPARATE SURPRISE FROM INVALIDITY

Use robust and contextual scores to prioritize review, then decide validity with market rules, source evidence, state, and corroboration. Preserve legitimate tail events.

3. TRACE THE FIRST BROKEN LAYER

Move from raw payload through normalization, eligibility, bars, features, and labels. Repairing the first divergence prevents inconsistent patches downstream.

4. PRESERVE CAUSAL KNOWLEDGE

Record event, arrival, revision, and effective times. A later correction creates a new vintage and must not be moved backward into an earlier decision.

5. MAKE EVERY REPAIR VISIBLE

Carry reason, original value, source IDs, method, uncertainty, boundaries, policy, and age. Synthetic values remain distinguishable from market observations.

6. VALIDATE BY CONSUMER

Execution, research, valuation, and risk need different freshness and fallback tolerances. Test shared definitions with consumer-specific action policies.

DECISION STANDARD

Promote only when expectedness, mechanism, evidence, causal timing, repair action, lineage, tests, monitoring, blast radius, fallback, and rollback are reproducible.

REFERENCE SHEET

Notation and quality-state contract

INTERMEDIATE

SYMBOL	OBJECT	MEANING
X_t	field value	Observed or derived value at decision index t
M_t	missing flag	Whether expected evidence is absent
R_t	reason code	Why a value is missing, invalid, or repaired
A_t	observation age	Decision time minus source update time
E_t	expected flag	Whether policy expects an observation
s_t	anomaly score	Degree of surprise under a chosen context
V_t	validity state	Valid, invalid, uncertain, or ineligible
P_t	price	Raw or adjusted price under stated convention
r_t	log return	log(P_t/P_(t-1)) on declared boundaries
tau	expiry limit	Maximum admissible age for a consumer

STATE BUNDLE

A safe analytical field is not just X_t. Preserve expectedness, observed status, source time, age, validity, reason, repair action, uncertainty, and policy so every consumer can interpret the value.

MINIMUM SNAPSHOT METADATA

Store source vintage, calendar, instrument master, schema, units, code, quality policy, thresholds, reason dictionary, repair lineage, tests, quality report, consumer compatibility, and snapshot hash.

PROCESS MAP

Evidence-to-publication quality workflow

INTERMEDIATE

1. **Declare expectations**
   Grain, identity, field, calendar, source, and cutoff
2. **Observe delivery**
   Sequence, heartbeat, latency, replay, and source status
3. **Validate structure**
   Schema, units, tick grid, bands, keys, and ordering
4. **Score context**
   Robust history, peers, state, and cross-source residuals
5. **Classify evidence**
   Valid tail, defect, missing mechanism, or uncertainty
6. **Choose action**
   Keep, expire, exclude, replace, impute, or quarantine
7. **Rebuild dependents**
   Bars, returns, features, labels, and model samples
8. **Validate causality**
   Vintage replay, fixtures, reconciliation, and parity
9. **Publish snapshot**
   Manifest, quality report, lineage, and compatibility
10. **Monitor and govern**
    Health tiers, incident queue, fallback, and rollback

Missing data is an observed state

INTERMEDIATE

ONE-SENTENCE MEANING

A missing value is an observation about the data-generating and delivery process, not merely an empty cell. Its reason, expectedness, and age determine whether it is harmless, informative, or dangerous.

MEMORY JOGGER

Ask why the value is absent before deciding how to fill it. Preserve value status beside value content so downstream code can distinguish no event, no delivery, invalid data, and not yet available.

WHY IT MATTERS IN QUANT RESEARCH

Missingness often correlates with illiquidity, stress, delisting, vendor coverage, or operational failure. A model can accidentally learn those mechanisms or acquire survivorship bias when rows are silently dropped.

FORMULA INTUITION

Represent each field as (X_t, M_t, R_t, A_t), where M is observed status, R is reason, and A is age. Estimate coverage with an explicit expected denominator rather than total wall-clock intervals.

SIMPLE MARKET EXAMPLE

An absent quote during a trading halt is structurally unavailable; an absent quote while peers update normally may indicate feed loss. Both display as null, but only the second supports packet recovery or source fallback.

PYTHON / SYSTEM IMPLEMENTATION

Use nullable values plus categorical reason codes, source timestamps, and expected flags. Feature builders should expose missing indicators and reject repairs that cross forbidden session, identity, or availability boundaries.

CONFUSIONS TO AVOID

Zero, unchanged, and missing are different states. Converting all three to one sentinel corrupts returns, liquidity, and model interpretation while making later diagnosis nearly impossible.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data-quality layer publishes value, status, reason, age, and policy action. Feature, backtest, execution, and risk services consume the full state rather than a visually clean scalar.

RELATED CONCEPTS AND QUICK RECALL

Related: missingness mechanism, data contract, observation age, eligibility. Recall task: Classify an absent quote during a halt and during an active market; choose a distinct system action for each.

Expected grids and coverage denominators

INTERMEDIATE

ONE-SENTENCE MEANING

An expected grid defines which instrument-time-field observations should exist under a declared calendar and source contract. Coverage is meaningful only when its numerator and denominator share that eligibility definition.

MEMORY JOGGER

Do not call a record missing until the system proves it was expected. Holidays, listing dates, session breaks, and source scope can remove observations from the denominator legitimately.

WHY IT MATTERS IN QUANT RESEARCH

Bad denominators create false outage alerts and hide real gaps. Coverage measured by symbol count can also overstate usable data when missing names carry most market capitalization or trading activity.

FORMULA INTUITION

Coverage_t = eligible observed records / eligible expected records. Publish count-, notional-, volume-, and universe-weighted versions when their economic interpretations differ.

SIMPLE MARKET EXAMPLE

A half-day equity session has 210 expected one-minute bars rather than 390. Penalizing the closed afternoon reports a 46 percent outage even when the feed is perfect.

PYTHON / SYSTEM IMPLEMENTATION

Generate expected rows from versioned calendars, listing intervals, venue hours, field frequency, and source entitlement. Join observations to the grid and retain missing reason plus denominator metadata.

CONFUSIONS TO AVOID

A regular time range is not an exchange calendar. Wall-clock grids fabricate gaps overnight, ignore auctions, and mishandle daylight-saving changes across markets.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calendar and instrument-master services create expected partitions before quality scoring. Dataset manifests store the exact expectation policy used for each coverage metric.

RELATED CONCEPTS AND QUICK RECALL

Related: trading calendar, eligibility, weighted coverage, listing interval. Recall task: Build the expected denominator for a one-minute stock dataset on an early-close day and explain which intervals are excluded.

MCAR, MAR, and MNAR in market data

INTERMEDIATE

ONE-SENTENCE MEANING

MCAR, MAR, and MNAR describe whether missingness is unrelated, conditionally explainable, or still dependent on the unseen value. They are assumptions about a mechanism, not labels proven by a null-pattern plot.

MEMORY JOGGER

Treat MCAR as a demanding hypothesis rather than a default. Market data disappear for reasons tied to liquidity, volatility, source capacity, security state, and the value itself.

WHY IT MATTERS IN QUANT RESEARCH

Imputation and inference can be biased when the mechanism is misclassified. Missing quotes during volatile events may remove the exact tail observations a risk model most needs.

FORMULA INTUITION

MCAR: $P(M|X,Z)=P(M)$. MAR: $P(M|X,Z)=P(M|Z)$ for observed Z. MNAR: missingness still depends on unseen X after conditioning; sensitivity analysis is then essential.

SIMPLE MARKET EXAMPLE

Thinly traded bonds quote less often when fair value is uncertain. Even after controlling for issue size and time of day, missingness may still relate to the unobserved executable spread.

PYTHON / SYSTEM IMPLEMENTATION

Model missing probability using only point-in-time observable drivers, compare results across missingness strata, and run pattern-mixture or bound analyses when MNAR is plausible. Document which conclusion depends on which assumption.

CONFUSIONS TO AVOID

A strong missingness predictor does not prove MAR because unseen values remain unavailable for testing. Adding future information to explain missingness also turns diagnosis into leakage.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Research validation stores mechanism assumptions with each repaired dataset. Risk review receives sensitivity results showing how conclusions change under adverse values for the unobserved records.

RELATED CONCEPTS AND QUICK RECALL

Related: selection model, pattern mixture, inverse probability weighting, sensitivity bound. Recall task: Give a market-data example for MCAR, MAR, and MNAR, then identify which assumption is least defensible.

MISSINGNESS FOUNDATIONS

Structural absence and no-trade intervals

INTERMEDIATE

ONE-SENTENCE MEANING

Structural absence occurs when the market process legitimately produces no event, such as no trade in a sparse interval or no quote during a suspension. It is different from failing to observe an event that should have been delivered.

MEMORY JOGGER

No event is information about activity, but it is not automatically a price observation. Preserve event count separately from any carried state.

WHY IT MATTERS IN QUANT RESEARCH

Treating no-trade bars as missing can discard liquidity information; treating them as fresh trades fabricates volume and price discovery. The correct representation depends on the consumer's question.

FORMULA INTUITION

For interval I, event_count(I)=0 is an observed fact when feed continuity is proven. A carried last price has source_time before I and must retain age rather than becoming a synthetic event at the interval close.

SIMPLE MARKET EXAMPLE

A small-cap stock has no trades from 10:04 to 10:08 while quotes update. A return study may carry an aged reference price, whereas a trade-arrival model uses four observed zero-count intervals.

PYTHON / SYSTEM IMPLEMENTATION

Publish separate columns for event count, last observed state, age, and interval eligibility. Bar builders should never infer a trade merely because they need a rectangular table.

CONFUSIONS TO AVOID

Forward-filled price does not mean continuous tradability. Converting an empty bar to volume zero and fresh close without status metadata mixes an observation with a derived state.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Event stores preserve sparse truth, while feature views create consumer-specific grids. Execution and liquidity features can reject aged states even when slower valuation processes accept them.

RELATED CONCEPTS AND QUICK RECALL

Related: event time, sparse process, carried state, zero-inflated count. Recall task: Explain how the same no-trade minute should appear in a return feature, volume feature, and execution simulator.

Feed outages and delivery gaps

INTERMEDIATE

ONE-SENTENCE MEANING

A delivery gap is a period when the source or transport path fails to provide expected records. It must be separated from a quiet market using sequence continuity, heartbeat state, peer sources, and later recovery evidence.

MEMORY JOGGER

Silence is ambiguous until the pipeline checks whether the market, source, or transport was silent. Operational metadata is part of market-data quality.

WHY IT MATTERS IN QUANT RESEARCH

Outages can create artificial calm, delayed jumps, stale spreads, and misleading labels. If volatile periods overload a feed, the missingness mechanism is directly related to risk.

FORMULA INTUITION

Gap length can be measured in missing sequence numbers, expected events, or elapsed decision time. Recovery completeness requires contiguous replay through the last committed offset, not merely a resumed heartbeat.

SIMPLE MARKET EXAMPLE

Venue A stops updating for 18 seconds while Venue B prices move and source heartbeats cease. When A resumes with a sequence jump, the interval should be quarantined until replay closes the missing range.

PYTHON / SYSTEM IMPLEMENTATION

Persist channel sequence, heartbeat, arrival latency, reconnects, delivery attempt, and replay status. Gate publication until the gap is filled or a consumer-approved degraded source is activated.

CONFUSIONS TO AVOID

A current timestamp after reconnection does not prove historical completeness. Forward-filling through the gap hides uncertainty and can make an outage look like low volatility.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Ingestion detects transport failure, normalization tracks recovered sequences, and dataset publication exposes partial windows. Production controls map severity to warn, degrade, stop, or fallback.

RELATED CONCEPTS AND QUICK RECALL

Related: sequence gap, heartbeat, replay, source failover. Recall task: List the evidence needed to distinguish an 18-second quiet market from an 18-second feed outage.

Censoring, truncation, and survivorship

INTERMEDIATE

ONE-SENTENCE MEANING

Censoring limits what is observed about a value, truncation excludes records outside a rule, and survivorship removes entities that no longer remain in the sample. Each mechanism changes the population represented by the dataset.

MEMORY JOGGER

Absence can be produced by the sampling design rather than a broken feed. Reconstruct inclusion rules before interpreting distributions or performance.

WHY IT MATTERS IN QUANT RESEARCH

Delisted securities, unfilled orders, minimum-reporting thresholds, and capped vendor histories can bias returns, costs, and failure rates. The cleanest-looking panel may describe only survivors.

FORMULA INTUITION

Right censoring records $T>c$ without observing exact T; truncation observes a row only if T passes an inclusion rule. Survivorship bias compares $E[Y|\text{survives}]$ with the desired $E[Y]$ over the original cohort.

SIMPLE MARKET EXAMPLE

A dataset containing today's constituents backfilled through history omits firms that failed. A strategy tested on this panel inherits better liquidity and returns than the investable universe available at each date.

PYTHON / SYSTEM IMPLEMENTATION

Store effective membership intervals, delisting returns, terminal states, and source retention rules. Build point-in-time cohorts and use survival-aware estimators when horizons end before outcomes are observed.

CONFUSIONS TO AVOID

Filling a missing delisting return with zero is not conservative; it can erase a large loss. Dropping unfilled orders also conditions cost analysis on successful execution.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The instrument master defines historical eligibility, while labels preserve censored outcomes and observation windows. Evaluation reports coverage by original cohort, not only final survivors.

RELATED CONCEPTS AND QUICK RECALL

Related: right censoring, left truncation, delisting bias, cohort. Recall task: Describe how current-constituent backfilling changes the target population of a historical strategy test.

Staleness is not the same as missingness

INTERMEDIATE

ONE-SENTENCE MEANING

A stale value exists but no longer represents the current decision state because its last update is too old for the use case. Missingness concerns absence; staleness concerns expired relevance.

MEMORY JOGGER

Every carried value needs a visible clock. A non-null cell can be less trustworthy than an explicit null when its source age is hidden.

WHY IT MATTERS IN QUANT RESEARCH

Stale quotes compress measured volatility, distort spreads, and create false cross-asset lead-lag. Execution systems require much shorter age limits than end-of-day valuation or slow fundamental models.

FORMULA INTUITION

age_t = decision_time - source_update_time. Eligibility requires age_t <= tau(instrument, field, session, regime, consumer), where tau is an explicit service-level threshold.

SIMPLE MARKET EXAMPLE

One leg of a pair updates at 10:00:05 while the other remains stamped 09:59:42. The apparent spread move may be asynchronous marking rather than an executable relative-value opportunity.

PYTHON / SYSTEM IMPLEMENTATION

Carry source time, receive time, age, update count, and expiry state through joins. Use as-of merges with maximum tolerance and report matched age distributions, not only row counts.

CONFUSIONS TO AVOID

A value does not become fresh when copied into a new timestamp. A single global tolerance ignores differences between liquid futures, thin bonds, auctions, and stress periods.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Feature services enforce consumer-specific age budgets before computing returns or spreads. Monitoring tracks stale notional and stale-duration tails by source and market state.

RELATED CONCEPTS AND QUICK RECALL

Related: as-of join, observation age, expiry, asynchronous prices. Recall task: Set different staleness rules for routing, intraday features, and end-of-day valuation; justify the ordering.

Cross-sectional missingness and selection bias

INTERMEDIATE

ONE-SENTENCE MEANING

Cross-sectional missingness occurs when fields or observations are unavailable for some instruments at a given decision date. If inclusion relates to size, liquidity, geography, or distress, complete-case analysis changes the investable population.

MEMORY JOGGER

Coverage must be profiled by economically important groups, not only averaged over symbols. A high headline rate can hide systematic loss of small or stressed names.

WHY IT MATTERS IN QUANT RESEARCH

Factor ranks, neutralization, and portfolio weights can shift when missing names are dropped. Coverage differences may masquerade as alpha or create unintended size and liquidity exposures.

FORMULA INTUITION

For group g, coverage_g = sum_i w_i 1(observed_i) / sum_i w_i over eligible i. Compare complete-case estimates with missing-indicator, bounded, and reweighted alternatives.

SIMPLE MARKET EXAMPLE

Analyst estimates cover 95 percent of large caps but 35 percent of microcaps. Ranking only observed names converts an earnings-revision signal into a large-cap-biased universe filter.

PYTHON / SYSTEM IMPLEMENTATION

Emit eligibility, observed status, group labels, and weight denominators. Fit any imputer or reweighting model inside training folds, then audit exposures before and after the missing-data policy.

CONFUSIONS TO AVOID

Median-filling the cross-section creates a synthetic neutral score but does not restore missing information. Dropping names after ranking changes both rank positions and portfolio constraints.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Universe construction publishes coverage slices and policy flags to feature ranking and portfolio construction. Risk attribution measures exposure introduced by availability rather than by the intended signal.

RELATED CONCEPTS AND QUICK RECALL

Related: complete-case analysis, availability bias, reweighting, universe construction. Recall task: Show how dropping uncovered microcaps can produce a hidden size tilt even when the feature itself is size-neutral.

An outlier is surprise, not proof of error

INTERMEDIATE

ONE-SENTENCE MEANING

An outlier is an observation far from a chosen reference distribution or neighborhood. It may be a data defect, a genuine market jump, a new regime, or a rare but valid tail event.

MEMORY JOGGER

Detection answers what is unusual; validation answers what is wrong. Never let a z-score make the truth decision by itself.

WHY IT MATTERS IN QUANT RESEARCH

Deleting genuine extremes understates tail risk and can manufacture stable strategies. Retaining defective points can dominate returns, labels, covariance, and risk estimates across many downstream windows.

FORMULA INTUITION

Define anomaly score $s(x|\text{context})$ and a separate validity decision V based on invariants, provenance, and corroboration. High s raises review priority but does not imply $V=0$.

SIMPLE MARKET EXAMPLE

A 14 percent earnings gap is extreme relative to recent daily returns but persists across venues and closes. A one-tick 90 percent drop that immediately reverses with no confirming quote is more likely erroneous.

PYTHON / SYSTEM IMPLEMENTATION

Store score, context window, rule evidence, source agreement, and final disposition separately. Recompute thresholds point in time and preserve all records in an audit view.

CONFUSIONS TO AVOID

Normality is not a validity rule, and frequency is not truth. Financial series are heavy-tailed, clustered, discontinuous, and subject to real structural breaks.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Exploratory analytics generates anomaly candidates; a governed validation service applies domain constraints and reason-coded actions. Risk uses valid tails even when feature models choose robust influence limits.

RELATED CONCEPTS AND QUICK RECALL

Related: anomaly score, validity rule, heavy tail, structural break. Recall task: Compare a confirmed earnings gap with an isolated decimal error and state which evidence changes the disposition.

Robust center and scale: median, MAD, and IQR

INTERMEDIATE

ONE-SENTENCE MEANING

Robust statistics estimate location and spread without allowing a few extreme observations to control the reference. Median, MAD, and IQR provide practical baselines for heavy-tailed market data.

MEMORY JOGGER

Make the reference resistant before measuring surprise. Classical mean and standard deviation can move toward the very point they are meant to flag.

WHY IT MATTERS IN QUANT RESEARCH

Stable anomaly scores improve triage for returns, spreads, volumes, and latency. Robustness reduces masking, but domain validation remains necessary because valid regimes can shift the entire distribution.

FORMULA INTUITION

MAD = median(|x_i - median(x)|); sigma_robust ~= 1.4826 MAD under a normal reference. IQR = Q_0.75 - Q_0.25, with fences commonly based on Q1 - k IQR and Q3 + k IQR.

SIMPLE MARKET EXAMPLE

Returns of 1, 0, -1, 1, and 40 percent have a mean pulled upward by the final point. Median and MAD keep the baseline near the ordinary observations and expose the jump clearly.

PYTHON / SYSTEM IMPLEMENTATION

Compute rolling robust statistics using only prior eligible observations, minimum counts, and explicit zero-scale handling. Partition by instrument, field, session phase, and regime when those contexts differ.

CONFUSIONS TO AVOID

MAD can be zero for discrete or unchanged series; dividing by it creates meaningless infinities. Robust does not mean immune to stale plateaus, mixed regimes, or coordinated source errors.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The detection layer publishes multiple scores and reference metadata. Consumers select thresholds by consequence, while the validation layer records whether the point was valid, repaired, or quarantined.

RELATED CONCEPTS AND QUICK RECALL

Related: median, MAD, IQR, influence function. Recall task: Calculate a robust score when MAD is positive, then specify a fallback for a zero-MAD quote series.

Detect in price space and return space

INTERMEDIATE

ONE-SENTENCE MEANING

Price-space checks enforce economic and venue constraints on levels, while return-space checks measure changes relative to time and context. Using both reveals errors that either view alone can miss.

MEMORY JOGGER

A valid level can produce an impossible jump, and a small return can still land off the tick grid. Match the diagnostic to the geometry of the defect.

WHY IT MATTERS IN QUANT RESEARCH

Decimal shifts, negative prices, stale plateaus, and incorrect adjustments propagate differently through derived features. Layered checks localize the defect before it contaminates rolling calculations.

FORMULA INTUITION

Level rules include P_t>0 where contract semantics require it and distance(P_t, tick_grid)<=epsilon. Return checks use r_t=log(P_t/P_(t-1)) with causal robust context and boundary awareness.

SIMPLE MARKET EXAMPLE

A stock print at 10.035 violates a 0.01 tick even if the return is small. A move from 100 to 10.03 lies on the tick grid but is suspicious in return space and cross-source context.

PYTHON / SYSTEM IMPLEMENTATION

Run structural level rules first, then temporal return rules, then cross-field and cross-source validation. Attach the earliest failed layer so repair logic does not confuse symptom with cause.

CONFUSIONS TO AVOID

Applying return thresholds across a split or session boundary creates false positives. Applying level bounds without instrument metadata can reject valid negative-settlement or special-market conventions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Normalization owns units, tick sizes, and adjustment state; anomaly detection owns contextual scores. Bar, feature, and label builders consume only records passing their declared policy.

RELATED CONCEPTS AND QUICK RECALL

Related: tick grid, log return, price band, adjustment state. Recall task: Give one defect caught only in level space and one caught mainly in return space; explain the rule order.

Multivariate and conditional anomalies

INTERMEDIATE

ONE-SENTENCE MEANING

A conditional anomaly looks unusual only after related fields, instruments, or market states are considered. Multivariate checks detect combinations that are individually plausible but jointly inconsistent.

MEMORY JOGGER

Ask whether the fields agree with one another, not merely whether each field is in range. Market data obey accounting, quote, and cross-asset relationships.

WHY IT MATTERS IN QUANT RESEARCH

Bad sizes, inverted quotes, currency mistakes, and asynchronous legs can pass univariate filters. Conditional diagnostics protect spreads, factors, covariance, and portfolio labels from coherent-looking single fields.

FORMULA INTUITION

Mahalanobis distance $d^2=(x-\mu)'\Sigma^{-1}(x-\mu)$ measures joint deviation under an elliptical reference. Robust or regularized covariance is required when dimensions are numerous, correlated, or heavy-tailed.

SIMPLE MARKET EXAMPLE

Bid and ask are both inside historical ranges, yet bid exceeds ask after one side becomes stale. The pair is invalid even though neither univariate value is extreme.

PYTHON / SYSTEM IMPLEMENTATION

Create cross-field invariants before statistical models, then use robust residuals from peer, factor, or state models. Log contributing dimensions so reviewers can explain the score.

CONFUSIONS TO AVOID

A large multivariate distance does not identify which field is wrong. Poorly conditioned covariance can amplify noise, and peer consensus can fail during broad market repricing.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Validation combines schema rules, quote geometry, reference-data consistency, and peer residuals. The quality report exposes both joint score and interpretable rule evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: Mahalanobis distance, robust covariance, cross-field invariant, peer residual. Recall task: Construct a case where bid and ask pass separate range checks but fail a joint validity rule.

Regime-aware and session-aware thresholds

INTERMEDIATE

ONE-SENTENCE MEANING

Regime-aware detection changes its reference for volatility, liquidity, session phase, or event state while keeping the decision rule causal. The goal is comparable surprise, not a threshold that relaxes after seeing the anomaly.

MEMORY JOGGER

Compare like with like. Opening auctions, lunch liquidity, crisis sessions, and overnight trading have different valid distributions.

WHY IT MATTERS IN QUANT RESEARCH

Static thresholds over-flag volatile regimes and under-flag quiet ones. A policy that adapts too quickly can also normalize a broken feed, so threshold changes need floors, caps, and monitoring.

FORMULA INTUITION

Use $s_t=|x_t-\text{center}_{(t-1,W,\text{state})}|/\text{scale}_{(t-1,W,\text{state})}$, with the window ending before t. Threshold k(state) is fixed by prior validation and consequence rather than fitted to the current point.

SIMPLE MARKET EXAMPLE

A 3 percent move may be routine around an earnings release but extraordinary for a short-dated Treasury future at midday. The context changes the anomaly score, not the need for source validation.

PYTHON / SYSTEM IMPLEMENTATION

Partition references by declared session phase and pre-known event flags, or use an online state model fitted only to prior data. Record the state assignment and fallback when samples are thin.

CONFUSIONS TO AVOID

Using full-sample volatility leaks future stress into earlier thresholds. Allowing every instrument-regime cell its own rule can overfit and eliminate meaningful comparability.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Research selects and validates state definitions; production computes them at decision time. Monitoring compares alert rates and false dispositions across states to detect policy drift.

RELATED CONCEPTS AND QUICK RECALL

Related: volatility regime, session phase, causal threshold, adaptive filter. Recall task: Explain how to score the same return differently at the open and midday without using future information.

OUTLIER DIAGNOSIS

Cross-source corroboration and consensus

INTERMEDIATE

ONE-SENTENCE MEANING

Cross-source corroboration compares independent observations after aligning identity, units, adjustment state, and timing. Agreement can validate a large move, while isolated disagreement can localize a defective source.

MEMORY JOGGER

Consensus is evidence only after semantics and clocks match. Three vendors that derive from one upstream feed are not three independent confirmations.

WHY IT MATTERS IN QUANT RESEARCH

Corroboration protects legitimate tail events from deletion and supports controlled fallback. It also measures source-specific reliability rather than assuming one vendor is always authoritative.

FORMULA INTUITION

Residual_i,t = source_i,t - robust_consensus_t, with tolerance scaled by tick size, spread, latency, and market state. Consensus should weight source independence and freshness, not only count.

SIMPLE MARKET EXAMPLE

One feed reports 52.10 while two independent venues and the national best quote remain near 25.10. A later correction confirms a transposed digit in the isolated source.

PYTHON / SYSTEM IMPLEMENTATION

Normalize all sources first, perform maximum-age joins, publish overlap counts, and retain source lineage. Use hierarchy rules for ties, stale consensus, and market-wide disagreement.

CONFUSIONS TO AVOID

A majority can be wrong when vendors share an origin or corporate-action table. Replacing a record with consensus without preserving the original creates unverifiable synthetic history.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The source-quality service scores disagreements and manages approved fallbacks. Dataset builders store original, replacement, consensus members, tolerance, and decision time.

RELATED CONCEPTS AND QUICK RECALL

Related: source independence, consensus price, fallback, reconciliation. Recall task: Name the checks required before treating two vendor prices as independent confirmation of a jump.

OUTLIER DIAGNOSIS

Corporate actions can imitate bad ticks

INTERMEDIATE

ONE-SENTENCE MEANING

Splits, dividends, spinoffs, symbol changes, and contract rolls create legitimate discontinuities or scale changes. A raw price jump becomes interpretable only after the action state and adjustment convention are known.

MEMORY JOGGER

Before deleting a large return, ask whether the economic unit changed. Adjustment errors often look cleaner than random bad ticks because every later value shares the new scale.

WHY IT MATTERS IN QUANT RESEARCH

Misapplied actions contaminate returns, volumes, market capitalization, factors, and labels over long windows. Double adjustment can be as damaging as no adjustment.

FORMULA INTUITION

For a split ratio q old shares per new share, compare theoretical ex-price $P_{ex}=P_{pre}/q$ under the declared convention. Total-return adjustment also includes cash distributions and action timing.

SIMPLE MARKET EXAMPLE

A 2-for-1 split takes an unadjusted price from about 100 to 50 while share count doubles. Cross-source persistence and action metadata support a valid unit change rather than a defective print.

PYTHON / SYSTEM IMPLEMENTATION

Maintain immutable raw prices and versioned action factors with announcement, effective, ex, and availability times. Validate factor continuity and rebuild adjusted views from source events.

CONFUSIONS TO AVOID

Backfilling a corrected action factor into all historical snapshots introduces knowledge-time leakage. Comparing adjusted and unadjusted sources without normalizing conventions creates false disagreement.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The reference-data service supplies point-in-time action state to normalization and return calculation. Quality incidents trace affected bars, features, models, and reports through snapshot lineage.

RELATED CONCEPTS AND QUICK RECALL

Related: split factor, total return, adjustment convention, contract roll. Recall task: Distinguish a genuine 2-for-1 split from a decimal error using persistence, metadata, volume, and source evidence.

Limit moves, halts, and auctions

INTERMEDIATE

ONE-SENTENCE MEANING

Market-state events change what prices, spreads, and update patterns are valid. Limit states, volatility pauses, auctions, and resumptions can create gaps or prints that violate ordinary continuous-session expectations.

MEMORY JOGGER

A rule without market-state awareness turns venue mechanics into defects. State transitions must travel with the market data they explain.

WHY IT MATTERS IN QUANT RESEARCH

Deleting auction or limit observations can erase price discovery and tail risk; mixing them with continuous trading can distort spreads and execution assumptions. Consumers need explicit eligibility by state.

FORMULA INTUITION

Validity is conditional: V(record|venue_state, session_phase, order_type, bands). Expected update intensity and price-band rules should be evaluated against the contemporaneous state.

SIMPLE MARKET EXAMPLE

After a volatility halt, the reopening auction prints far from the last continuous trade and at large size. The print is valid but may be excluded from a continuous-spread estimator.

PYTHON / SYSTEM IMPLEMENTATION

Ingest venue status messages, auction indicators, bands, and session labels. Preserve all events, then publish state-specific views with documented inclusion rules.

CONFUSIONS TO AVOID

A persistent price after a limit lock is not necessarily stale in the ordinary sense. Conversely, a halt flag arriving late cannot retroactively justify decisions that lacked that information.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Market-state normalization precedes anomaly classification, bar construction, features, and execution simulation. Backtests use the state known at each decision cutoff.

RELATED CONCEPTS AND QUICK RECALL

Related: volatility halt, price limit, auction, session state. Recall task: Explain why a reopening-auction print may be valid for risk history but ineligible for a continuous-spread feature.

OUTLIER DIAGNOSIS

Volume, size, and quote anomalies

INTERMEDIATE

ONE-SENTENCE MEANING

Outliers occur in quantity, notional, depth, spread, and update rate as well as price. Field-specific semantics and conservation checks often detect size errors that return filters cannot see.

MEMORY JOGGER

A plausible price with an impossible size can still destroy liquidity and impact estimates. Validate units, multipliers, signs, and aggregation paths before statistical surprise.

WHY IT MATTERS IN QUANT RESEARCH

A 100x volume error biases participation rates, capacity, slippage, VWAP, and bar weighting. Quote-depth duplication can make a market appear liquid precisely when the feed is replaying.

FORMULA INTUITION

Notional = price * quantity * contract_multiplier * FX conversion. Reconcile aggregate volume to eligible unique events and use robust log-scale scores for heavy-tailed positive quantities.

SIMPLE MARKET EXAMPLE

A futures feed changes size from contracts to underlying units while prices remain normal. Volume jumps by the contract multiplier and passes simple positivity checks.

PYTHON / SYSTEM IMPLEMENTATION

Attach units and multipliers to schemas, deduplicate by event identity, and compare count, quantity, and notional across source and bar layers. Flag sudden changes in unit ratios.

CONFUSIONS TO AVOID

Winsorizing size can hide a multiplier error without repairing its economic meaning. Large block trades are valid tails and require venue and event-type context.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Normalization validates quantity semantics; bar builders reconcile totals; cost models consume only unit-certified fields. Monitoring tracks price-size ratios and source-specific distribution shifts.

RELATED CONCEPTS AND QUICK RECALL

Related: contract multiplier, notional, depth, volume reconciliation. Recall task: Describe how a unit switch can pass price checks yet multiply estimated capacity by 100.

The anatomy of a bad tick

INTERMEDIATE

ONE-SENTENCE MEANING

A bad tick is a delivered market event whose price, size, identity, time, or state is inconsistent with authoritative rules or corroborating evidence. The defect may originate at venue, vendor, transport, parser, normalization, or join layers.

MEMORY JOGGER

Do not repair the symptom until the failure layer is located. The same spike can come from a true source event, a decimal parser, a duplicated message, or an incorrect instrument mapping.

WHY IT MATTERS IN QUANT RESEARCH

One bad event can contaminate OHLC bars, returns, volatility, labels, stops, and rolling features for many periods. Root-cause classification determines how broadly the system must rebuild.

FORMULA INTUITION

Model the path raw payload -> normalized event -> eligible event -> bar -> feature. At each edge verify identity, units, sequence, policy, and reconciliation so the first divergence is observable.

SIMPLE MARKET EXAMPLE

A raw payload contains 10.03, while independent sources remain near 100.30. If the normalized event also shows 10.03, the defect predates bar construction; if raw shows 100.30, parsing is responsible.

PYTHON / SYSTEM IMPLEMENTATION

Retain raw bytes, canonical fields, transform versions, source IDs, and rule outcomes. Generate a defect graph linking every downstream artifact that used the event.

CONFUSIONS TO AVOID

Calling every suspicious return a bad tick skips causal diagnosis. Editing only the final bar leaves contaminated trade counts, volume, and dependent features elsewhere.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Incident tooling moves from record evidence to lineage impact and deterministic replay. Publication creates a new immutable snapshot rather than overwriting the disputed history.

RELATED CONCEPTS AND QUICK RECALL

Related: root cause, event lineage, parser defect, impact graph. Recall task: Trace an anomalous bar backward and name the evidence that distinguishes source error from parser error.

Hard invariants: prices, ticks, bands, and identities

INTERMEDIATE

ONE-SENTENCE MEANING

Hard invariants encode conditions that must hold under known instrument and venue semantics. They catch impossible states before probabilistic methods evaluate whether a possible state is merely unusual.

MEMORY JOGGER

Apply certainty before surprise. Positivity, tick grid, price bands, timestamp order, quote geometry, and key validity produce explainable failures with low ambiguity.

WHY IT MATTERS IN QUANT RESEARCH

Invariant failures usually justify immediate quarantine and prevent propagation. They also expose reference-data defects when many records fail the same supposedly stable rule.

FORMULA INTUITION

Grid error = |P/tick - round(P/tick)|; accept within numeric tolerance. Additional rules include effective identity interval, venue band inclusion, finite values, and bid<=ask where the market convention requires it.

SIMPLE MARKET EXAMPLE

A print of 25.007 fails a 0.01 tick grid even though its return is tiny. A price of 25.01 may pass the grid but fail the active venue band after a limit-state update.

PYTHON / SYSTEM IMPLEMENTATION

Compile invariants from versioned reference data and evaluate them at event time plus knowledge time. Store rule ID, inputs, tolerance, policy version, and exact failure reason.

CONFUSIONS TO AVOID

Hard rules are only hard when metadata is correct and contemporaneous. Floating-point equality without tolerance creates false failures, while stale tick-size tables create systematic misclassification.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Normalization blocks impossible canonical events and alerts reference-data owners when failure clusters indicate metadata drift. Downstream consumers never need to rediscover basic semantic violations.

RELATED CONCEPTS AND QUICK RECALL

Related: tick size, price band, quote geometry, identity interval. Recall task: Write a tick-grid validation rule and specify the metadata and numeric tolerance it requires.

Isolated spikes and immediate reversals

INTERMEDIATE

ONE-SENTENCE MEANING

An isolated spike is a sharp move that lacks persistence or corroboration and is followed by a near-immediate reversal. The pattern is strong defect evidence but must be tested against auctions, news, and genuinely thin trading.

MEMORY JOGGER

A bad tick often leaves a V-shaped footprint because only one event carries the wrong scale. Real repricing usually changes subsequent observable state, though exceptions exist.

WHY IT MATTERS IN QUANT RESEARCH

Spike-reversal patterns can dominate realized volatility and trigger false stops or labels. Automatic deletion remains risky because flash moves and busted trades require distinct handling.

FORMULA INTUITION

Define forward-backward residuals $a=|\log(P_t/P_{t-1})|$ and $b=|\log(P_{t+1}/P_t)|$ plus recovery $c=|\log(P_{t+1}/P_{t-1})|$. Large a and b with small c raises suspicion but uses future evidence only for offline validation.

SIMPLE MARKET EXAMPLE

Prices 100.00, 10.03, and 100.05 create two enormous returns but almost no net move. Quotes and other venues staying near 100 strengthen the decimal-error explanation.

PYTHON / SYSTEM IMPLEMENTATION

Use the pattern to flag records after the next event arrives, never to grant historical real-time knowledge. For causal feature generation, quarantine the uncertain interval until policy or independent evidence resolves it.

CONFUSIONS TO AVOID

The reversal test is not available at the time of the spike and cannot be inserted into a backtest as if known. Illiquid instruments may legitimately gap between sparse prints without a smooth intermediate path.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Offline quality processing can revise a later dataset vintage, while real-time systems use conservative fallback or stop logic. Snapshot metadata preserves which version each decision used.

RELATED CONCEPTS AND QUICK RECALL

Related: V-shaped anomaly, busted trade, delayed confirmation, data vintage. Recall task: Explain which parts of a spike-reversal diagnosis are available immediately and which arrive only later.

Duplicates, cancels, corrections, and event identity

INTERMEDIATE

ONE-SENTENCE MEANING

Market feeds may replay an event, cancel a prior event, or publish a corrected version. Correct handling requires economic event identity and lifecycle semantics rather than dropping visually identical rows.

MEMORY JOGGER

Equal values can be two trades; unequal values can be one corrected trade. Identity and message type decide the lifecycle.

WHY IT MATTERS IN QUANT RESEARCH

Replay duplicates inflate volume and counts, while ignored cancels preserve trades that no longer stand. Over-aggressive deduplication can erase legitimate repeated executions at the same price and size.

FORMULA INTUITION

Canonical state is a function of source, venue, channel, sequence, event ID, message type, and correction link. Idempotence requires applying the same delivery twice to produce no second economic effect.

SIMPLE MARKET EXAMPLE

Two consecutive 100-share trades at 25.10 are valid if event IDs differ. A cancel referencing the first event removes it from active trade views while preserving both messages in the audit log.

PYTHON / SYSTEM IMPLEMENTATION

Persist transport identity and economic identity separately, model event lifecycle explicitly, and use deterministic winner rules. Rebuild affected bars from active events instead of subtracting ad hoc aggregates.

CONFUSIONS TO AVOID

Dropping duplicates over all columns is not a business rule. Deleting canceled history destroys the evidence needed to reconstruct what was known before the cancel arrived.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Ingestion handles replay idempotence, normalization resolves lifecycles, and point-in-time views expose the active state at each cutoff. Bar lineage references the exact event versions included.

RELATED CONCEPTS AND QUICK RECALL

Related: idempotence, replay, cancel, correction chain. Recall task: Explain why row equality is insufficient for deduplication and how a late cancel changes historical vintages.

Bar contamination and deterministic rebuilding

INTERMEDIATE

ONE-SENTENCE MEANING

A contaminated bar is an aggregate whose OHLC, volume, count, or derived statistics include defective or ineligible events. Repair should rebuild the entire bar from governed source events, not patch only the visible field.

MEMORY JOGGER

Aggregates are consequences, not evidence. Fix the event set and let every bar field follow from the same deterministic policy.

WHY IT MATTERS IN QUANT RESEARCH

Changing only a bad close can leave an impossible high, inflated volume, wrong VWAP, and inconsistent trade count. Downstream features then disagree about what happened.

FORMULA INTUITION

OHLCV = Aggregate({e_i: eligible(e_i, policy, cutoff)}). Reconciliation checks low<=min(open,close)<=max(open,close)<=high, volume=sum(size_i), and count=number of active eligible trades.

SIMPLE MARKET EXAMPLE

One decimal-error trade becomes the bar low and close while adding 500 shares. Removing its price but not its event leaves volume and VWAP contaminated.

PYTHON / SYSTEM IMPLEMENTATION

Store event-to-bar membership, total order, eligibility reasons, and policy version. Recompute bars and every dependent rolling window, then compare hashes outside the impacted range.

CONFUSIONS TO AVOID

Patching OHLC fields manually can create a bar no possible event path produced. Rebuilding without the original ordering can change open and close nondeterministically.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The bar service is a reproducible projection of the event store. Lineage allows incident response to identify precisely which features, labels, backtests, and live decisions require refresh.

RELATED CONCEPTS AND QUICK RECALL

Related: OHLCV invariant, VWAP, event membership, deterministic replay. Recall task: List every bar field that one bad trade can affect and describe the correct rebuild boundary.

Forward fill: persistent state with an expiry

INTERMEDIATE

ONE-SENTENCE MEANING

Forward fill carries a previously observed state into later rows without creating a new observation. It is defensible only when the variable is state-like, the consumer permits it, and an explicit age limit prevents false freshness.

MEMORY JOGGER

Carry the value, never its timestamp or certainty. A filled cell should announce its origin and age.

WHY IT MATTERS IN QUANT RESEARCH

Forward fill can align sparse reference data or quotes for bounded uses, but it suppresses volatility and invents availability through outages when applied carelessly. The damage is greatest around session and regime boundaries.

FORMULA INTUITION

$X^*_t = X_s$ for latest $s \le t$ only if $age = t - s \le \tau$ and no forbidden boundary lies between s and t. Also emit observed_t=0, source_time=s, age, and fill_reason.

SIMPLE MARKET EXAMPLE

A daily sector classification may persist until a new effective record. An executable quote should expire quickly and must not be carried across a halt, overnight boundary, or source outage.

PYTHON / SYSTEM IMPLEMENTATION

Use as-of joins with maximum tolerance, partition by instrument and field, and reset at identity, session, and action boundaries. Unit tests should perturb future values and confirm earlier fills do not change.

CONFUSIONS TO AVOID

Forward fill is not imputation of a new measurement and should not be counted as coverage. Filling both price and volume can fabricate trades and artificial zero returns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Reference-data joins and some slow features may use governed carries; routing and risk apply stricter expiry. Dataset reports separate observed, carried, expired, and missing proportions.

RELATED CONCEPTS AND QUICK RECALL

Related: as-of join, state variable, expiry, synthetic observation. Recall task: Give one field suitable for long carry and one requiring short expiry; identify the boundary resets for each.

Interpolation and model-based imputation

INTERMEDIATE

ONE-SENTENCE MEANING

Interpolation and model-based imputation create estimates for unobserved values using neighboring or related data. They are analytical constructs with uncertainty, not recovered market observations.

MEMORY JOGGER

If a method invents a value, label it synthetic and name every information source used. Offline smoothness is not causal availability.

WHY IT MATTERS IN QUANT RESEARCH

Imputation can stabilize exploratory panels and some slow models, but it distorts realized paths, extremes, and execution if treated as observed. Multiple imputation or uncertainty bounds may be more honest than one precise fill.

FORMULA INTUITION

Linear interpolation is $X(t)=X_a+(X_b-X_a)(t-t_a)/(t_b-t_a)$. For decisions before t_b it uses future information; model-based estimates require predictive distributions and fitted-parameter lineage.

SIMPLE MARKET EXAMPLE

Interpolating a missing daily risk-free rate between published endpoints may be acceptable for retrospective attribution. Interpolating an intraday trade price creates a path no participant could necessarily transact.

PYTHON / SYSTEM IMPLEMENTATION

Fit imputers inside training folds, restrict features to point-in-time inputs, emit method and uncertainty, and maintain an observed-only validation slice. Compare conclusions under no-fill and adverse-fill scenarios.

CONFUSIONS TO AVOID

A visually smooth series can be statistically wrong and economically impossible. Single imputation understates uncertainty and can cause models to overtrust fabricated precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Offline research views may offer explicitly synthetic variants, while backtest execution and labels default to observed evidence. Model registry entries bind the exact imputation artifact and training cutoff.

RELATED CONCEPTS AND QUICK RECALL

Related: linear interpolation, multiple imputation, predictive distribution, leakage. Recall task: State when interpolation becomes future leakage and how an observed-only holdout can reveal imputer dependence.

Cross-source replacement and authoritative fallback

INTERMEDIATE

ONE-SENTENCE MEANING

Cross-source replacement substitutes a disputed value with an aligned observation from an approved source. It is defensible only when semantic equivalence, timing, authority, and uncertainty are documented.

MEMORY JOGGER

Replacement is a lineage edge, not an overwrite. Preserve the original, replacement, reason, and source-ranking decision.

WHY IT MATTERS IN QUANT RESEARCH

Fallback can restore continuity during vendor failure, but subtle convention differences can create artificial jumps or mixed histories. A lower-latency source may also be less final or less complete.

FORMULA INTUITION

Accept replacement Y_t for X_t when identity, field, unit, adjustment, venue scope, and age all match within policy. Reconciliation residual $|X-Y|$ must be interpreted using tick, spread, and state-dependent tolerance.

SIMPLE MARKET EXAMPLE

A primary consolidated close is missing, while an approved exchange close with the same auction scope and final status is available. The fallback is acceptable but remains labeled by its actual source.

PYTHON / SYSTEM IMPLEMENTATION

Maintain a versioned source hierarchy and compatibility matrix. Emit fallback flag, original status, replacement ID, matched age, residual, and policy version into the curated row.

CONFUSIONS TO AVOID

Two fields with the same name may use different session, adjustment, currency, or finality conventions. Blending vendors without regime controls can make strategy performance depend on source switching.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A source-quality service activates degraded modes and publishes compatibility metadata. Consumers can reject fallback records when their decision requires the primary source's specific semantics.

RELATED CONCEPTS AND QUICK RECALL

Related: source hierarchy, semantic equivalence, degraded mode, compatibility matrix. Recall task: List the equivalence checks required before using an exchange close to replace a missing consolidated close.

Robust transforms and tail influence

INTERMEDIATE

ONE-SENTENCE MEANING

Winsorization caps values at chosen quantiles, clipping enforces fixed bounds, and robust transforms reduce the influence of extremes. These are modeling choices, not evidence that the original observations were wrong.

MEMORY JOGGER

Control influence in the feature layer while preserving truth in the data layer. A capped tail must remain traceable to its uncapped value.

WHY IT MATTERS IN QUANT RESEARCH

Robust transforms can prevent one valid extreme from dominating a model, but they also change tail information, ranks, moments, and portfolio response. Thresholds must be fitted and validated causally.

FORMULA INTUITION

Winsorized $x^*=\min(\max(x,q_{low}),q_{high})$; a signed log transform can be $\mathrm{sign}(x)\log(1+|x|/c)$. Store bounds, training sample, affected rate, and original x.

SIMPLE MARKET EXAMPLE

A valid one-day 40 percent return may be capped for a linear cross-sectional model while remaining unchanged in risk and event studies. The cap limits model leverage without declaring the event erroneous.

PYTHON / SYSTEM IMPLEMENTATION

Fit bounds inside each training fold, apply them unchanged to validation data, and emit cap indicators. Compare raw, transformed, and rank-based features across regimes and portfolio tails.

CONFUSIONS TO AVOID

Clipping bad data can hide a unit or parser defect and let it pass silently. Full-sample quantiles leak future distribution shifts and make historical transformations irreproducible.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Feature engineering owns influence transforms; quality control owns validity decisions. The feature registry records raw lineage, transform artifact, fitted cutoff, and downstream consumers.

RELATED CONCEPTS AND QUICK RECALL

Related: winsorization, clipping, robust loss, feature influence. Recall task: Explain why a valid extreme may be capped for one model but must remain intact for risk estimation.

Causal repair and knowledge-time integrity

INTERMEDIATE

ONE-SENTENCE MEANING

Causal repair uses only evidence available by the historical decision cutoff, while later corrections create new data vintages. A final cleaned history is not automatically valid for backtesting earlier decisions.

MEMORY JOGGER

Separate what ultimately became true from what the system could know then. Both histories are useful, but they answer different questions.

WHY IT MATTERS IN QUANT RESEARCH

Late cancels, revised actions, and vendor backfills can improve final accuracy while introducing look-ahead if applied retroactively. Point-in-time evaluation must replay the uncertainty and provisional states actually faced.

FORMULA INTUITION

A record is eligible when available_time<=cutoff and effective_from<=cutoff<effective_to under the active revision at that cutoff. Later evidence may change only later snapshots.

SIMPLE MARKET EXAMPLE

A bad trade is canceled ten minutes after it entered a bar. A decision made in minute two saw the provisional bar; an end-of-day research snapshot can exclude the canceled event.

PYTHON / SYSTEM IMPLEMENTATION

Store event, arrival, publication, effective, and revision times. Build as-of views by data vintage and run future-perturbation tests that require earlier outputs to remain unchanged.

CONFUSIONS TO AVOID

Sorting corrected records by event time can move future knowledge backward. A pristine final dataset can overstate the reliability of a live strategy that operated through unresolved anomalies.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Snapshot services expose real-time, provisional, and final views with explicit contracts. Backtests choose the view matching the intended operational decision and document revision latency.

RELATED CONCEPTS AND QUICK RECALL

Related: bitemporal data, as-of snapshot, revision latency, look-ahead bias. Recall task: Describe how a late trade cancel should appear in a minute-two backtest and in a final end-of-day dataset.

Reason codes, provenance, and reversible actions

INTERMEDIATE

ONE-SENTENCE MEANING

Every missingness classification, anomaly decision, and repair should produce a stable reason code linked to source evidence and policy version. Reversibility means the original record and transformation path remain available.

MEMORY JOGGER

A clean value without an explanation is an undocumented opinion. The audit trail should reconstruct both what changed and why it was allowed.

WHY IT MATTERS IN QUANT RESEARCH

Reason-coded lineage enables incident analysis, policy comparison, targeted rebuilds, and model impact assessment. It also prevents different teams from silently assigning different meanings to the same null.

FORMULA INTUITION

Represent action as output_id -> {source_ids, rule_ids, scores, decision, old_value, new_value, policy_id, run_id}. Aggregate quality metrics from these atomic outcomes.

SIMPLE MARKET EXAMPLE

A quote is expired because age exceeds 500 ms during continuous trading. The curated null links to the original quote, source timestamp, age rule, market state, and consuming policy.

PYTHON / SYSTEM IMPLEMENTATION

Use controlled vocabularies with machine code and human description, plus append-only action logs. Validate that every changed or excluded record has exactly one terminal disposition.

CONFUSIONS TO AVOID

Free-text notes alone cannot support reliable metrics or automated replay. A single generic OUTLIER code hides whether the cause was tick grid, stale state, source disagreement, or confirmed real jump.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The catalog exposes rule and action lineage to feature, model, backtest, and production artifacts. Policy upgrades can compare changed dispositions before promotion.

RELATED CONCEPTS AND QUICK RECALL

Related: audit trail, lineage edge, policy version, disposition. Recall task: Specify the minimum provenance fields for a forward-filled quote and for an excluded bad trade.

Quarantine and human review

INTERMEDIATE

ONE-SENTENCE MEANING

Quarantine isolates records whose materiality is high and evidence is insufficient for automatic acceptance or repair. Human review should resolve defined questions with preserved context, deadlines, and reproducible dispositions.

MEMORY JOGGER

Uncertainty is a valid system state. Do not force ambiguous evidence into clean or bad merely to keep a pipeline visually complete.

WHY IT MATTERS IN QUANT RESEARCH

Quarantine prevents uncertain records from contaminating sensitive consumers while allowing investigation and controlled degraded operation. Review capacity should focus on economically material and decision-relevant cases.

FORMULA INTUITION

Priority can combine affected notional, consumer severity, anomaly confidence, duration, and blast radius. Service levels define when the system must fallback, stop, or publish partial data.

SIMPLE MARKET EXAMPLE

A large-cap closing auction price disagrees across sources near portfolio rebalance. The record enters high-priority review because valuation, risk, and benchmark tracking are exposed.

PYTHON / SYSTEM IMPLEMENTATION

Build evidence packets with raw payloads, neighboring events, source comparisons, reference state, fired rules, lineage, and proposed actions. Require reviewer identity and structured resolution code.

CONFUSIONS TO AVOID

Manual review must not become an untracked spreadsheet edit. Low-value alerts can exhaust reviewers and delay the few incidents that materially affect trading or risk.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The incident queue integrates quality scores, consumer lineage, owner routing, and publication gates. Resolved cases become fixtures and policy-training examples.

RELATED CONCEPTS AND QUICK RECALL

Related: quarantine, materiality, incident queue, service-level objective. Recall task: Design a priority score for an ambiguous close and explain how its affected consumers change urgency.

Blast radius and downstream impact

INTERMEDIATE

ONE-SENTENCE MEANING

Blast radius is the set of datasets, features, labels, models, backtests, reports, and live decisions affected by a defective source record or policy. It depends on lineage and temporal propagation, not only row count.

MEMORY JOGGER

One bad tick can travel through many windows. Scope the first bad input, every derived artifact, and the last output whose calculation depends on it.

WHY IT MATTERS IN QUANT RESEARCH

Targeted impact analysis avoids both under-repair and needless full-history rebuilds. It also tells owners which published research conclusions or production decisions require revalidation.

FORMULA INTUITION

For rolling window W, a defect at t can affect features through t+W-1; recursive models may carry influence longer. Dependency graphs add cross-sectional ranks, labels, and portfolio aggregates.

SIMPLE MARKET EXAMPLE

A false daily close changes next-day return, 20-day volatility for 20 sessions, cross-sectional ranks on each affected day, and any model retrained from those samples.

PYTHON / SYSTEM IMPLEMENTATION

Maintain record- and partition-level lineage, window metadata, model training manifests, and consumer subscriptions. Compute impacted ranges and verify unchanged hashes outside them after rebuild.

CONFUSIONS TO AVOID

Rebuilding only the source date misses lagged and rolling consequences. Recomputing everything without scope increases operational risk and makes incident validation harder.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Incident response queries the dependency catalog before publication. Owners receive affected artifact IDs, ranges, severity, required action, and replacement snapshot.

RELATED CONCEPTS AND QUICK RECALL

Related: data lineage, dependency graph, rolling window, artifact manifest. Recall task: Map the downstream range of one bad close through one-day return, 20-day volatility, and monthly model retraining.

Golden fixtures and adversarial tests

INTERMEDIATE

ONE-SENTENCE MEANING

Golden fixtures are small, versioned datasets with known expected outcomes for difficult defects and valid extremes. Adversarial tests deliberately inject failures to prove that controls catch defects without deleting real market behavior.

MEMORY JOGGER

Test the pair: one observation the rule must reject and one similar observation it must preserve. A detector is incomplete if it only demonstrates sensitivity.

WHY IT MATTERS IN QUANT RESEARCH

Fixtures prevent policy regressions, reveal order dependence, and document institutional knowledge about actions, halts, duplicates, gaps, and bad ticks. They make unusual cases reproducible during code review.

FORMULA INTUITION

A good test asserts record disposition, reason, repaired value, lineage, affected aggregates, and idempotent replay. Property tests cover invariants across broad generated inputs.

SIMPLE MARKET EXAMPLE

Inject a decimal-error spike and a confirmed earnings gap of similar magnitude. The first should be quarantined or excluded; the second should remain valid with tail status.

PYTHON / SYSTEM IMPLEMENTATION

Create fixtures for DST, early closes, splits, late cancels, sequence gaps, stale joins, zero MAD, source disagreement, and auction transitions. Pin expected manifests and output hashes.

CONFUSIONS TO AVOID

Tests using only ordinary days prove little about quality systems. Snapshot tests that accept any changed output by regeneration can bless a defect rather than detect it.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Continuous integration runs fast fixtures on every change and larger replay suites before policy promotion. Production incidents contribute minimized cases after review.

RELATED CONCEPTS AND QUICK RECALL

Related: golden dataset, property test, fault injection, regression test. Recall task: Design a two-case test that rejects a decimal error while preserving a similarly large confirmed market jump.

Monitoring missingness, outliers, and source health

INTERMEDIATE

ONE-SENTENCE MEANING

Production monitoring tracks quality dimensions with explicit denominators, expected ranges, and consumer-aware actions. Rates should be segmented by source, instrument group, field, session, and market state.

MEMORY JOGGER

Monitor the mechanism, not only the cleaned output. A stable final table can hide rising fallback, carry, and quarantine rates.

WHY IT MATTERS IN QUANT RESEARCH

Early detection limits contamination and reveals slow source degradation. Multiple metrics separate coverage loss, latency, disagreement, anomaly drift, and repair dependence.

FORMULA INTUITION

Key measures include expected coverage, stale notional, sequence gaps, anomaly candidate rate, confirmed defect rate, source residual, fallback share, quarantine age, and revision impact.

SIMPLE MARKET EXAMPLE

Observed coverage stays at 99.9 percent while forward-fill share rises from 2 to 30 percent. A value-only dashboard appears healthy, but freshness monitoring reveals source failure.

PYTHON / SYSTEM IMPLEMENTATION

Publish counts and economic denominators, compare to seasonal baselines, and map thresholds to warn, degrade, stop, or rollback. Alert on change and duration, not a single noisy point.

CONFUSIONS TO AVOID

A single composite quality score can average away a critical stop condition. Alert thresholds trained on crisis data may normalize dangerous behavior during normal markets.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Source operations owns transport metrics; data quality owns semantic metrics; consumer teams own tolerances. A shared scorecard records the action contract for each threshold.

RELATED CONCEPTS AND QUICK RECALL

Related: data observability, stale notional, fallback share, quality SLO. Recall task: Choose five metrics that would expose a hidden outage even when the final table has no nulls.

Consumer-specific quality policies

INTERMEDIATE

ONE-SENTENCE MEANING

A quality policy defines which observed, stale, missing, fallback, or repaired states a particular consumer may use. Validity is not universal because execution, research, valuation, and risk have different latency and uncertainty tolerances.

MEMORY JOGGER

The same record can be admissible for audit and forbidden for routing. Name the decision before naming the repair.

WHY IT MATTERS IN QUANT RESEARCH

Shared raw evidence supports consistency, while consumer policies prevent one permissive cleaning rule from silently weakening high-stakes systems. Policies also make degraded behavior explicit and testable.

FORMULA INTUITION

Eligibility = f(record_state, consumer, horizon, cutoff, market_state, tolerance). The function returns pass, degraded, stop, or quarantine with a reason and fallback path.

SIMPLE MARKET EXAMPLE

A five-minute-old bond quote may support approximate end-of-day marking but not an executable intraday spread. A vendor fallback may be fine for research and unacceptable for a regulated benchmark.

PYTHON / SYSTEM IMPLEMENTATION

Version policy matrices by consumer and field, test them against shared fixtures, and publish policy ID with each dataset. Do not materialize one ambiguous clean flag for every downstream service.

CONFUSIONS TO AVOID

Consumer-specific does not mean arbitrary team-specific behavior. Policies should share definitions, evidence, governance, and minimum controls while varying only justified tolerances and actions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data contract registry maps quality states to feature, label, backtest, execution, valuation, and risk consumers. Compatibility checks block undeclared state use.

RELATED CONCEPTS AND QUICK RECALL

Related: quality policy, degraded mode, admissibility, consumer contract. Recall task: Compare acceptable age and fallback states for execution, research, and valuation; explain each difference.

Promotion, rollback, and reproducible snapshots

INTERMEDIATE

ONE-SENTENCE MEANING

Promotion publishes a validated immutable snapshot that binds data, code, schema, policy, source cutoffs, quality report, and lineage. Rollback selects a prior compatible snapshot rather than attempting to undo edits in place.

MEMORY JOGGER

Never publish latest without identity. A reproducible dataset is a named artifact, not a mutable folder.

WHY IT MATTERS IN QUANT RESEARCH

Snapshot discipline allows research results to be reproduced, policy changes to be compared, and incidents to be reversed safely. Compatibility checks prevent rollback to structurally valid but semantically incompatible data.

FORMULA INTUITION

Snapshot ID hashes ordered input checksums, schema IDs, policy, code, parameters, source vintages, and quality outcomes. Promotion requires declared gates and atomic publication.

SIMPLE MARKET EXAMPLE

A new outlier policy unexpectedly removes legitimate auction prints. Consumers switch back to the previous approved snapshot while the policy is corrected and replayed in isolation.

PYTHON / SYSTEM IMPLEMENTATION

Build candidate snapshots, run fixtures and reconciliation, compare disposition diffs, obtain approvals, and publish manifests atomically. Retain prior snapshots and documented migration paths.

CONFUSIONS TO AVOID

Overwriting partitions destroys comparison and rollback. A byte-identical file is not semantically identical if its schema, policy, calendar, or source vintage metadata changed.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dataset registry serves approved versions to research and production. Deployment systems verify consumer compatibility and expose current, previous, and candidate quality reports.

RELATED CONCEPTS AND QUICK RECALL

Related: immutable snapshot, manifest, promotion gate, rollback. Recall task: List the manifest fields required to reproduce a repaired dataset and safely roll back its policy.

PROCESS DIAGRAM

Evidence-preserving defect pipeline

INTERMEDIATE

1. **Raw delivery**
   Payload, source, sequence, receive time, checksum
2. **Canonical event**
   Identity, units, venue state, corrections, clocks
3. **Expected evidence**
   Calendar, listing, entitlement, field frequency
4. **Validation stack**
   Hard invariants, robust context, source agreement
5. **Disposition**
   Keep, expire, exclude, replace, estimate, quarantine
6. **Deterministic rebuild**
   Events to bars, features, labels, and samples
7. **Quality gates**
   Reconciliation, fixtures, causality, compatibility
8. **Immutable publish**
   Snapshot, report, lineage, monitor, rollback

CONTROL PRINCIPLE

Evidence is append-only; interpretations are versioned. A repaired output remains linked to its original record, validation evidence, consumer policy, and historical knowledge cutoff.

HEATMAP

Missingness patterns reveal mechanisms

INTERMEDIATE

	Open	Midday	Close	Overnight	Stress	Revision
Liquid equity	FULL	FULL	FULL	N/E	PART	FULL
Microcap	PART	LOW	PART	N/E	LOW	FULL
Halted name	FULL	N/E	N/E	N/E	N/E	FULL
Feed B	FULL	LOW	PART	N/E	N/E	PART
New listing	N/E	N/E	PART	N/E	PART	FULL
Delisted name	PART	PART	N/E	N/E	N/E	LOW
Slow fundamental	LOW	LOW	LOW	LOW	LOW	FULL

INTERPRETATION

Rows with similar null rates can represent illiquidity, a halt, source failure, listing history, terminal disappearance, or slow publication. The pattern proposes a mechanism; clocks, state, and source evidence decide it.

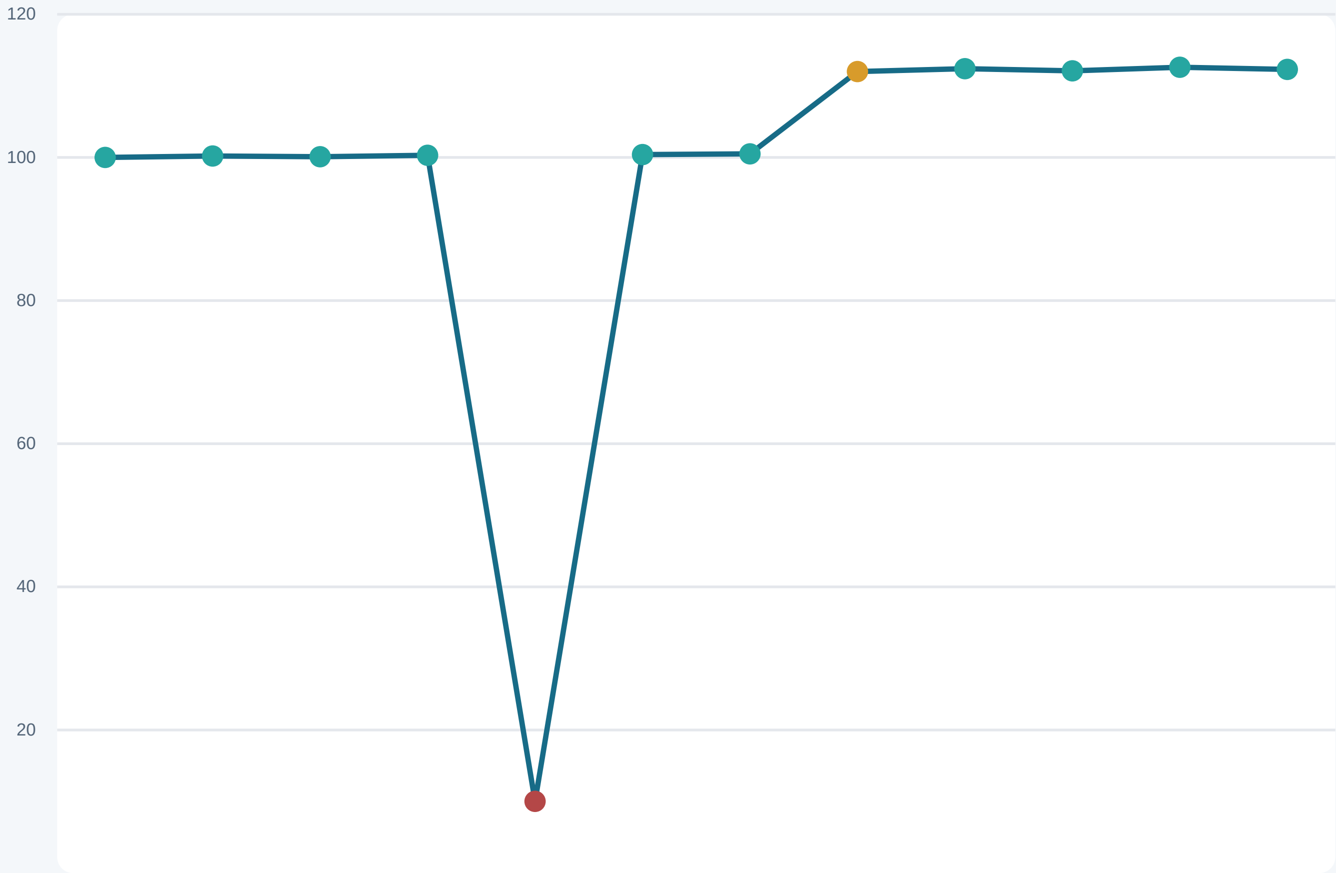
DENOMINATOR RULE

N/E means not expected and must not reduce coverage. Report count-, notional-, and universe-weighted coverage because missing distressed or small names changes conclusions differently.

ANNOTATED CHART

Bad tick versus genuine repricing

INTERMEDIATE



ISOLATED DECIMAL-SCALE EVENT

The 10.03 print immediately reverses, disagrees with contemporaneous quotes, and violates the source's prevailing scale. Offline evidence strongly supports quarantine and event-level rebuild.

PERSISTENT MARKET REPRICING

The move toward 112 receives subsequent confirmations and remains the new state. Its robust score is large, yet persistence and cross-source evidence require preserving the tail event.

EVIDENCE STACK

Combine tick rules, bands, sequence health, source age, venue state, corporate actions, independent sources, and later correction messages. Magnitude alone cannot determine truth.

DECISION TREE

Decision tree for a suspicious observation

INTERMEDIATE

TIMING WARNING

Later reversal and correction evidence may resolve an offline vintage, but a historical decision cannot use evidence that had not arrived. Preserve provisional and final states separately.

COMPARISON TABLE

Repair methods and their evidence burden

INTERMEDIATE

ACTION	DEFENSIBLE WHEN	PRIMARY RISK	TRACE REQUIRED
Keep	Valid unusual event	Tail influence	Score + corroboration
Expire	State exists but too old	Coverage loss	Source time + age rule
Exclude	Record invalid for consumer	Selection bias	Original + reason + scope
Forward fill	State persists within age limit	False freshness	Origin + age + boundary
Interpolate	Offline smooth field	Future leakage	Endpoints + uncertainty
Replace	Approved equivalent source	Semantic mismatch	Both sources + hierarchy
Winsorize	Model influence needs control	Destroyed tail shape	Raw + fitted bounds
Quarantine	Evidence material and uncertain	Delayed publication	Packet + owner + SLA

SELECTION RULE

Use the least assumptive action compatible with the consumer. Every created value remains synthetic; every excluded value remains available; and every policy decision is reproducible at its historical cutoff.

WORKED CALCULATION

Worked calculation: robust score and bar impact

INTERMEDIATE

OBSERVED RETURNS

Prior eligible returns are -0.3%, -0.1%, 0.0%, 0.1%, 0.2%, 0.3%, and 0.4%. A new event implies -89.97%, so magnitude clearly warrants investigation.

ROBUST CENTER

The prior median is 0.1%. Absolute deviations are 0.4, 0.2, 0.1, 0.0, 0.1, 0.2, and 0.3 percentage points; MAD is 0.2%.

ROBUST SCALE

Normal-consistent scale is 1.4826 x 0.2% = 0.2965%. The robust score is about |-89.97%-0.1%|/0.2965% = 304, an extreme triage signal.

VALIDITY EVIDENCE

Price 10.03 still lies on a one-cent grid, so grid validity alone passes. Quotes and two independent sources remain near 100.30, and the next trade is 100.35.

DISPOSITION

Policy classifies the event as a source-scale defect and excludes it from the regular active-trade view. The raw message remains immutable with score and evidence.

BAR IMPACT

Before repair, low and close are 10.03 and the bad size contributes 500 shares. Event-level rebuild restores low near 100, close 100.35, volume minus 500, and corrected VWAP and count.

PROPAGATION

The defect affects the current bar, next return, rolling volatility windows, labels using the close, cross-sectional ranks, and any model sample containing those features.

CAUSAL CAVEAT

The next-trade confirmation was unavailable at event time. A real-time strategy must model provisional state, quarantine latency, or source fallback rather than assume the final repair.

VALIDATION

Replay the raw sequence twice, require identical canonical state, reconcile rebuilt bars to active events, and assert unchanged hashes outside the impacted dependency range.

FORMULA SHEET

Formula and notation reference

INTERMEDIATE

COVERAGE

coverage = observed eligible / expected eligible. Publish the expectation policy and count-, notional-, volume-, or universe-weighted denominator.

OBSERVATION AGE

$A_t = \text{decision\_time} - \text{source\_update\_time}$. Eligibility requires $A_t <= tau$ for the instrument, field, session, state, and consumer.

ROBUST SCALE

$MAD = median(|x_i - median(x)|)$; sigma_robust is approximately 1.4826 MAD under a normal reference. Define minimum samples and zero-scale fallback.

CAUSAL ANOMALY SCORE

$s_t = |x_t - center_{(t-1,W)}| / scale_{(t-1,W)}$. The reference ends before t and resets at declared identity, action, or session boundaries.

LOG RETURN

$r_t = \log(P_t/P_{(t-1)})$. Verify adjustment convention, positive-price domain, source freshness, and legal boundary before interpreting magnitude.

TICK-GRID ERROR

$e_{grid} = |P/tick - round(P/tick)|$. Compare with a numeric tolerance using the contemporaneous tick schedule and price band.

MULTIVARIATE DISTANCE

$d^2 = (x-mu)' \, Sigma^{-1} (x-mu)$. Use robust or regularized covariance and report which dimensions contributed to the alert.

LINEAR INTERPOLATION

$X(t)=X_a+(X_b-X_a)(t-t_a)/(t_b-t_a)$. Endpoint b is future information for any decision before it became available.

SNAPSHOT IDENTITY

Hash ordered source checksums, schema, calendar, reference data, code, policy, parameters, repair log, quality report, and consumer compatibility.

DIAGNOSTIC TABLE

Symptoms, likely causes, and first checks

INTERMEDIATE

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Smooth prices during volatility	Hidden forward fill	Age and observed flag	Expire stale state
One-bar 90% drop	Decimal or identity error	Raw payload and peers	Quarantine event
Persistent 50% level shift	Split or unit change	Action metadata	Rebuild convention
Volume rises 100x	Multiplier or duplicate	Units and event IDs	Normalize or dedupe
Negative spread	Stale side or bad join	Leg timestamps	As-of with expiry
Coverage falls in microcaps	Selection mechanism	Group denominators	Audit universe bias
Outliers vanish in stress	Adaptive threshold drift	State and floors	Freeze policy bounds
History changes weekly	Mutable vendor vintage	Snapshot diffs	Bind immutable manifest
Bar fields disagree	Patched aggregate	Event membership	Rebuild all OHLCV

TRIAGE ORDER

Protect sensitive consumers, freeze publication, preserve evidence, identify first divergence, verify clocks and units, scope the blast radius, rebuild deterministically, reconcile, and publish a new snapshot.

TIMELINE

Provisional evidence and later correction

INTERMEDIATE

READING RULE

Track what the system knew at each cutoff. The suspicious trade, peer disagreement, official cancel, and final corrected bar arrive at different times, so they belong to different data vintages and decision states.

DECISION AT 10:00:00.5

The suspicious trade is known but the later peer evidence and cancel are not. A live policy may quarantine, fallback, reduce risk, or proceed provisionally.

DECISION AT 10:00:05

Independent disagreement is now available and can strengthen source-specific suspicion. The official cancel still cannot be assumed.

AFTER THE CANCEL

The active event lifecycle changes. New snapshots exclude the canceled trade from final bars while preserving earlier provisional vintages.

VALIDATION TEST

Modify all records strictly after an earlier cutoff and recompute. Features and quality decisions before that cutoff must remain identical.

OPERATIONAL LESSON

Backtests should model the intended vintage, decision latency, fallback, and unresolved-state behavior rather than use only final cleaned history.

HEATMAP

Policy stability across regimes and consumers

INTERMEDIATE

	Quiet	Volatile	Auction	Illiquid	Execution	Research
Tick-grid rules	ROBUST	ROBUST	ROBUST	ROBUST	ROBUST	ROBUST
Robust return score	ROBUST	WARN	WEAK	WARN	WARN	ROBUST
Quote expiry	ROBUST	WARN	WEAK	WEAK	ROBUST	WARN
Source consensus	ROBUST	WARN	WEAK	WARN	ROBUST	ROBUST
Forward fill	WARN	WEAK	STOP	WEAK	STOP	WARN
Winsor bounds	ROBUST	WARN	WARN	WARN	WEAK	ROBUST
Quarantine SLA	ROBUST	ROBUST	WARN	WARN	ROBUST	WARN

READING THE MAP

Hard semantic rules remain stable, while statistical thresholds and fills need state-specific validation. A STOP cell means the method is incompatible with that context, not merely lower scoring.

PROMOTION STANDARD

Require acceptable behavior in every material regime and consumer cell. Document fallback for thin samples, monitor disposition drift, and reject policies that gain coverage by hiding uncertainty.

PROMOTION GATE

Dataset promotion checklist

INTERMEDIATE

EXPECTATION CONTRACT

Grain, key, listing interval, calendar, source entitlement, field frequency, and expected denominator are declared and versioned.

EVIDENCE PRESERVATION

Raw payloads, source identity, sequence, clocks, reference state, original values, and checksums are immutable and queryable.

VALIDATION LAYERS

Schema, units, tick grid, bands, lifecycle, robust context, peer residuals, actions, and market state produce interpretable outcomes.

MISSINGNESS POLICY

Structural absence, outage, not-yet-available, invalid, censored, stale, and uncovered states have distinct reasons and consumer actions.

REPAIR TRACE

Every exclusion, fill, estimate, cap, or replacement preserves original evidence, method, uncertainty, boundaries, policy, and source time.

CAUSAL INTEGRITY

As-of eligibility, revision vintages, future perturbation, fold-local fitted transforms, and historical decision latency are validated.

REBUILD AND RECONCILE

Events reproduce bars; bars reproduce returns; dependent windows are scoped; hashes remain unchanged outside impacted ranges.

TESTS AND REVIEW

Golden fixtures, valid-tail controls, replay idempotence, zero-scale cases, outages, actions, auctions, and compatibility tests pass.

PUBLICATION AND OPERATIONS

Manifest, quality report, lineage, source cutoffs, owner, alerts, fallback, rollback, and consumer compatibility are atomically bound.

FINAL REVIEW

Final active recall and system index

INTERMEDIATE

CLASSIFY BEFORE FILLING

Expectedness and reason come before repair. No event, no delivery, invalid data, stale state, censoring, and unavailable-yet are distinct mechanisms.

PRESERVE REAL TAILS

Anomaly scores measure surprise. Rules, state, source independence, corporate actions, persistence, and lifecycle evidence decide validity.

REPAIR THE FIRST DIVERGENCE

Trace raw payload to event, eligibility, bar, feature, label, and model. Rebuild downstream artifacts from governed evidence.

CARRY THE CLOCK

Source time, arrival, availability, effective intervals, revisions, age, and decision cutoff determine causal eligibility.

LABEL SYNTHETIC VALUES

Forward fills, interpolation, replacements, and caps preserve their origins, uncertainty, boundaries, policy, and original values.

MEASURE WHAT IS HIDDEN

Monitor expected coverage, stale notional, sequence gaps, fallback share, quarantine age, disagreement, revision impact, and disposition drift.

GOVERN BY CONSUMER

Execution, research, valuation, and risk share evidence and definitions but apply justified age, fallback, and stop tolerances.

PUBLISH IMMUTABLY

A reproducible snapshot binds sources, schema, calendars, code, policy, repair log, tests, quality report, lineage, compatibility, and hash.

FINAL RECALL

For any repaired value, explain why it was expected, what failed, which evidence was available, what action occurred, what changed downstream, and how to roll back.